R13

Code No: 126AP

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
DISTRIBUTED SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

JNTUH Used papers 2023

PART - A

(25 Marks)

1.a) Define authentication. [2]
b) Define and explain about the distributed systems. [3]
c) What reconfiguration would you expect to occur in the NTP synchronization subnet? [2]
d) What is an election algorithm? [3]
e) Write the characteristics of inter process communications. [2]
f) Differentiate between unicast and multicast communication. [3]
g) List the characteristics of distributed file systems. [2]
h) What are the requirements of the distributed file systems? [3]
i) Write the comparisons of methods for concurrency control. [2]
j) What is deadlock? Give the characteristics of a deadlock condition to hold in a distributed system. [3]

PART - B

(50 Marks)

2. Explain in brief about system models of distributed systems. [10]

OR

3.a) Explain about architectural elements.
b) Write short notes on characteristics of distributed systems. [5+5]

4.a) Devise a protocol for basic multicast over IP multicast.
b) Is it possible to implement either a reliable or an unreliable failure detector using an unreliable communication channel? [5+5]

OR

5.a) Discuss about global states.
b) Discuss about distributed mutual exclusion. [5+5]

6. Discuss about the API for the Internet protocols. [10]

OR

7. What is a Remote Procedure Call(RPC)? Explain. [10]

8. Why should UFIDs be unique across all possible file system? How is uniqueness for UFIDs ensured? Explain. [10]

OR

9. Explain about design and implementation issues of distributed shared memory. [10]

10.a) Discuss about timestamp ordering.

b) Discuss the concurrency control in distributed transactions. [5+5]

OR

11. Explain about flat and nested distributed transactions. [10]

---ooOoo---

JNTUH Used papers 2023